Notes on Phonological Feature Representations

This document describes the 19-feature binary representations that defines all the phonemes used for modeling this version of the kidwords dataset.

1. Theory Notes

The features synthesizes classical generative distinctive feature theory with some modern pieces to create a clean, non-contradictory vector space for modeling in artificial neural networks (ANNs):

- Classical Distinctive Features (Chomsky & Halle, 1968 - SPE): foundational principles of binary categorical oppositions across stricture, voicing, and resonance. Key features include [±syllabic], [±sonorant], [±consonantal], [±continuant], [±delayed release], [±voice], [±nasal], [±lateral], [±strident], [±high], [±low], [±tense], and [±anterior].
- Unified Articulator Place Geometry (Clements, 1985; Clements & Hume, 1995): Resolves an incompatibility between consonant place features (e.g., anterior, coronal) and vowel coordinates (e.g., high, back) by organizing place under active physical articulators: [LABIAL], [CORONAL], and [DORSAL]. This allows consonants and vowels to be encoded within the exact same dimensional sub-space without dormant or NaN coordinates.
- Distributed Representations in PDP styling: Distributed reps are used for phonology.

2. Design

A. Resolving Affricate Representation: Stricture vs. Release

In initial designs, modeling affricates (e.g., /tʃ/ in *hutch* or /dʒ/ in *judge*) as simultaneously [+plosive, +fricative] creates both theoretical and computational contradictions:

- Stricture Contradiction: Oral stops and fricatives have mutually exclusive radical strictures (complete central closure vs. continuous turbulent airflow without closure).
- Vector Superposition: Activating both plosive and fricative units simultaneously signals a consonant cluster (e.g., /ts/ or /t-ʃ/) rather than an atomic segment occupying a single temporal slot, distorting the attractor landscape.

The Solution: Plosives and fricatives remain mutually exclusive. Affricates pattern underlyingly as oral stops in primary stricture (plosive=1, fricative=0, continuant=0). The turbulent friction phase of the release is encoded via two distinct, complementary features:

- [strident=1]: Captures high-frequency acoustic turbulence generated against the sharp obstruction of the teeth/palate, shared with sibilant fricatives (/s, z, ʃ, ʒ/).

- [delayed_release=1]: Explicitly models the slow, non-instantaneous release phase of affricates (/tʃ, dʒ/), clearly distinguishing them from simple instantaneous plosives (/t, d, p, b, k, g/).

B. Eliminating Glide-Vowel problems via [syllabic]

In standard distinctive feature frameworks, glides (/j, w/) and high vocoids (/ɪ, ʊ/) share the identical major class profile (sonorant=1, consonantal=0, continuant=1) and place specifications (palatal glide /j/ matches front vocoid /ɪ/; labial-velar glide /w/ matches back rounded vocoid /ʊ/). Without a syllabicity distinction, these pairs form exact identical duplicates in a 16-feature space.

The Solution: Incorporating the canonical major-class feature [syllabic] resolves this collision cleanly:

- syllabic=1: Assigned strictly to all vocalic syllable nuclei (all 13 monophthongs and diphthongs).
- syllabic=0: Assigned to all consonantal margins, including the vocoid glides /w/ and /j/ (y). This produces a distinct minimal pair where I vs. y and U vs. w differ solely by syllabic status, reflecting their exact linguistic distribution.

C. Explicit Liquid Separation via [lateral]

While the liquid consonants /l/ and /r/ differ in coronal anteriority (/l/ is [+anterior], /r/ is [-anterior]), adding the binary feature [lateral] (1 for /l/, 0 for /r/ and all other phonemes) prevents the neural network from relying on a single subtle place distinction for two manners that exhibit distinct acoustic profiles and phonotactic branching constraints.

D. Unified Binary Place Across Consonants and Vowels

Clements and Hume's (1995) articulator-based place hierarchy is represented without ternary or continuous medial values:

- [labial]: Lip involvement. Active for bilabials/labiodentals (/p, b, m, f, v/), the glide /w/, rounded vowels (/uː, ʊ, oʊ/), and labial-gliding diphthongs (W /aʊ/, O /ɔɪ/).
- [coronal]: Constriction formed by the tongue blade, tip, or front. Active for dental, alveolar, and post-alveolar consonants (/t, d, s, z, θ, ð, ʃ, ʒ, tʃ, dʒ, n, l, r/), the palatal glide /j/ (y), front vowels (/iː, ɪ, eɪ, ɛ, æ/), and front-gliding diphthongs (A /aɪ/, O /ɔɪ/).
- [dorsal]: Constriction formed by the tongue dorsum. Active for velars (/k, g, w/) and back/central vowels (/uː, ʊ, oʊ, ɑː, ʌ/).
- Binary Height: Mid vowels (/eɪ, ɛ, oʊ/) are cleanly specified without ternary logic as [high=0, low=0], preserving linear separability for connectionist units.

3. The Definitive 19-Feature Binary Matrix (phonreps_enhanced.csv)

The table below details all 36 phonemic symbols used in the corpus across 19 binary distinctive features (1 = positive/present, 0 = negative/absent). Empty alignment positions (_) are set to all zeros (19 zeros).

4. Neural Network Pipeline Integration

1. Phonological Output Vectors (phon.csv): Each of the 7 temporal slots (Onset: slots 0–2; Nucleus: slot 3; Coda: slots 4–6) is encoded as a 19-element binary vector. The full word representation is the flattened concatenation of all 7 slots ($7 \times 19 = 133$ units per word). Empty alignment positions (_) are set to all zeros.
2. Orthographic Input Vectors (orth.csv): Each of the 10 letter slots (Onset: slots 0–3; Vowel Nucleus: slots 4–5; Coda: slots 6–9) is encoded as a 26-dimensional one-hot binary vector ('a' through 'z'). Empty alignment positions (_) are set to 26 zeros. This yields an input layer of $10 \times 26 = 260$ binary units per word.
3. Attractor Topography: Because stricture, vocal tract articulators, voicing, delayed release, laterality, and syllabicity are all represented explicitly, the network develops natural attractor basins reflecting human phonological neighborhood density and phonotactic constraints.

References

Chomsky, N., & Halle, M. (1968). *The Sound Pattern of English*. New York: Harper & Row.

Clements, G. N. (1985). The geometry of phonological features. *Phonology Yearbook*, 2, 225–252.

Clements, G. N., & Hume, E. V. (1995). The internal organization of speech sounds. In J. A. Goldsmith (Ed.), *The Handbook of Phonological Theory* (pp. 245–306). Oxford: Blackwell.

Harm, M. W., & Seidenberg, M. S. (1999). Phonology, reading acquisition, and dyslexia: Insights from connectionist models. *Psychological Review*, 106(3), 491–528.

Harm, M. W., & Seidenberg, M. S. (2004). Computing the meanings of words in reading: Cooperative division of labor between visual and phonological processes. *Psychological Review*, 111(3), 662–720.

Plaut, D. C., McClelland, J. L., Seidenberg, M. S., & Patterson, K. (1996). Understanding normal and impaired word reading: Computational principles in quasi-regular domains. *Psychological Review*, 103(1), 56–115.

Sagey, E. (1986). *The Representation of Features and Relations in Non-Linear Phonology* (Doctoral dissertation, MIT).

Seidenberg, M. S., & McClelland, J. L. (1989). A distributed, developmental model of word recognition and naming. *Psychological Review*, 96(4), 523–568.